ABDULLAH ALI

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Education

University of Illinois at Chicago — Dec 2024

B.S. Computer Science — 3.9 GPA

- **Relevant Coursework**: Artificial Intelligence, Data Science, Linear Algebra, Statistics, Linguistics, Database Systems, Data Structures, Algorithms, Object Oriented Programming, Software Engineering, Systems Programming
- **Certifications/Awards:** AWS Certified Cloud Practitioner Certification, Magna Cum Laude, Chancellor's Fellows

Skills

- **Languages:** Python, SQL, Swift, Go, Javascript, Typescript, HTML, CSS
- **AI/ML Frameworks:** LangChain, LangGraph, RAG, Pinecone, OpenAI API, TensorFlow, PyTorch, scikit-learn
- **Tools:** FastAPI, React, Next.js, PostgreSQL, Spark, Supabase, Docker, AWS, Pandas, Numpy, Git, Tableau

Work Experience

Software Engineer | *ICQC Online* — Dec 2024 - Current

- Elevated revenue by $10,000 by collaborating with cross-functional board members and introducing an online donation feature developed in React, resulting in increased donor retention and broader community participation
- Built an internal AI agent employing OpenAI and LangChain that enables teachers to retrieve student insights and generate reports through a natural language chat platform, reducing manual report generation time by 65%
- Designed a PostgreSQL schema for managing data across 800+ students, teachers, and classes, implementing role-based access control to protect sensitive academic records and enforce secure access for authenticated users
- Managed version control via Git and applied CI/CD pipelines with GitHub Actions for automated deployment

AI Engineer Intern | *Argonne National Laboratory* — May 2024 - Dec 2024

- Boosted workflow efficiency by 46% by developing a research-tuned AI agent leveraging automated tool calling and RAG to include unstructured data in the knowledge base, using Pinecone to create and store vector embeddings
- Designed scalable agentic system utilizing OpenAI and Llama APIs with Python to power natural language interactions, supporting dynamic content generation, supercomputer task automation, and contextual queries
- Evolved agent interaction by implementing hand gesture and speech recognition, and by creating an agentic workflow to predict user intentions, automate processes, and validate output by prompt engineering the LLM
- Presented application and insights to audience of 500+ at Sandia XR Conference and UIC AI Impact Day

Software Developer | *Electronic Visualization Laboratory* — Sep 2023 - May 2024

- Analyzed effectiveness of pedestrian safety measures by tracking vehicle speeds and sudden stops, leveraging TensorFlow for vehicle detection, PyTorch for speed classification, and Tableau to visualize speed distributions
- Engineered a data pipeline to extract insights from unstructured data by storing data in AWS S3, processing it with Spark for feature extraction, and modeling structured results in a MariaDB relational database
- Reduced latency by 50% by integrating computer-embedded sensors to enable real-time edge computing

Machine Learning Engineer Intern | *University of Illinois at Chicago* — Jun 2022 - Aug 2022

- Enhanced accuracy of bone age regression model by 27% by engineering Python algorithm to standardize clinical data
- Accomplished a 10x reduction in dataset standardization time by automating manual process of rotating x-ray images to normalize spinal alignment, employing computer vision using OpenCV to get outlines of objects in images
- Conducted in-depth analysis of model performance by testing the model using k-fold Cross Validation

Projects

AI Travel Agent | *LangGraph, Python, FastAPI, Next.js* — July 2025

- Consolidated travel planning tasks into a LangGraph workflow equipped to summarize real-time flight data, scrape reddit for personalized suggestions, and create detailed itineraries automatically added to the user's Google Calendar
- Applied LangGraph for granular control and tracking of agent logic and streamed response tokens to reduce latency

Tweet Political Party Classifier API | *Python, AWS, Docker, scikit-learn, NLTK* — May 2025

- Deployed a machine learning classifier achieving 83% accuracy in predicting political party affiliation from raw tweets using NLP techniques as a REST API on AWS using API Gateway and Lambda with a Docker image on ECS

Real Estate Investment Model | *Python, Pandas, scikit-learn, numpy, Seaborn* — Dec 2024

- Enabled data-driven real estate decisions by constructing a Random Forest Regression machine learning model to identify optimal investment neighborhoods in Chicago, achieving an R-squared score of 0.85 and an MAE of 12.5%